EXPERIMENT - 4

Data Modelling and Form Design in Zoho Creator

Library Book Issue Management

Application Name	Library Book Issue Management
Form Name	Book_Issue_Request_Form
Report Name	Book_Issue_Request_Report
Student Name	CALEB D
Student Roll No.	241401016
Experiment	4
Platform	Zoho Creator
Record Date	09 October 2026

Lab Record	Design of a structured library request form using all thirteen Basic Field types, four logical sections, validation rules, and submitted report records.

Department of Computer Science and Business Systems

Rajalakshmi Engineering College

1. Objective

To design and build Book_Issue_Request_Form in Zoho Creator using all thirteen Basic Fields specified in the library project, and to store valid requests as structured records for library staff.

2. Problem Statement

A physical library register and verbal requests make pending requests, availability checks and collection schedules difficult to track. A single online form is required to capture member identity, contact details, book category, borrowing purpose, collection schedule and acknowledgement of library rules.

3. Scope of the Use Case

Item	Implementation
Application	Library Book Issue Management
Primary users	Students, faculty and staff members; library staff
Form	Book_Issue_Request_Form
Report	Book_Issue_Request_Report
Processing objective	Review requests, verify availability and prepare copies for collection

4. Business Data Model

The core entity is a book issue request. Each submitted form creates one record, identified by the Creator record ID. Member details, contacts, request details and scheduling information belong to that record. Request_Status, Staff_Notes and Confirmation_Sent support later staff processing.

Data flow: Member input > validation > request record > staff review > availability confirmation > collection.

241401016

5. Basic Field Mapping - All 13 Fields

#	Basic Field	Exact Field Name	Purpose / example
1	Name	Member_Name	First and last name of member
2	Email	Email_Address	Recipient for availability confirmation
3	Address	Hostel_Residential_Address	Hostel or home address
4	Phone	Contact_Number	Indian mobile number with +91
5	Single Line	Register_Roll_Number	Student register or employee number
6	Multi Line	Purpose_Of_Borrowing	Detailed borrowing purpose
7	Number	Number_Of_Copies	Request between 1 and 5 copies
8	Date	Preferred_Issue_Date	Preferred collection date
9	Time	Preferred_Collection_Time	Preferred collection time
10	Drop Down	Book_Category	Computer Science / Business / General
11	Radio	Member_Type	Student / Faculty / Staff
12	Multi Select	Subject_Areas_Of_Interest	AI / Data Science / Machine Learning
13	Checkbox	Agree_To_Library_Rules	Acceptance of library rules

6. Section-wise Form Design

Section	Fields
A - Member Details	Member_Name; Register_Roll_Number; Member_Type
B - Contact Details	Email_Address; Contact_Number; Hostel_Residential_Address
C - Book Request Details	Book_Category; Subject_Areas_Of_Interest; Number_Of_Copies; Purpose_Of_Borrowing
D - Scheduling & Acknowledgement	Preferred_Issue_Date; Preferred_Collection_Time; Agree_To_Library_Rules

241401016

7. Validation Rules

Field / group	Configuration
Member_Name	Mandatory; first and last name sub-fields.
Contact_Number	Mandatory; India +91. Deluge requires a ten-digit mobile number beginning 6, 7, 8 or 9 after country code 91.
Register_Roll_Number	Mandatory single-line identity reference.
Member_Type; Book_Category	Mandatory single choice in each field.
Email_Address	Built-in email format validation; required before setting Available.
Number_Of_Copies	Default 1; validation rejects blank, zero and values outside 1-5.
Preferred_Issue_Date	New and rescheduled requests cannot have a past date. An unchanged historical date is retained during staff processing.
Agree_To_Library_Rules	Mandatory acknowledgement; unchecked submission is rejected.

8. Implementation Procedure

1. Created the application and the named form manually in Zoho Creator.

2. Added the thirteen Basic Fields with the exact field names from the project PDF.

3. Grouped the fields into the four specified sections and configured category, member type and subject choices.

4. Enabled mandatory fields, email formatting, Indian phone format, consent and Deluge range/date checks.

5. Added staff-only processing controls and generated Book_Issue_Request_Report.

6. Submitted sample requests and verified saved records and invalid-input rejection.

241401016

9. Sample Business Records

The following three fictitious requests were saved and verified in Development on 09 October 2026. They satisfy the project requirement for at least three submitted sample records.

Member / reference	Request details	Saved status
Priya Nair LAB260901	Student; Computer Science & Engineering; 2 copies; 10-Oct-2026	Available
Aarav Shah QA-FAC-002	Faculty; Business & Management; 5 copies; 12-Oct-2026	Pending
Divya Krishnan QA-STF-003	Staff; General Reference; 1 copy; 13-Oct-2026	Under Review

The approved Priya email test used the owner mailbox. Other mock addresses use example.com, which the notification function deliberately skips.

10. Evidence Guide

The next pages show genuine Creator screenshots: the builder across its scroll positions, the live form, at least three saved records, and a rejected invalid phone number. No sample-document watermark or sample loan-application screenshot is used.

Builder screenshots are from Development; live-form screenshots are from Production v1.1. Preview values are unsaved and do not represent an additional submitted request. Report and validation screenshots are identified in their captions.

Workflow summary

1. Member opens the form and supplies the required details.
2. Creator validates the submitted values and saves a request with Pending status.
3. Library staff review and filter the report, then set the processing status.
4. An Available request with a usable email triggers the configured confirmation integration.
5. The member collects the book at the agreed time; staff can mark the request Collected.

241401016

10 - Implementation Evidence

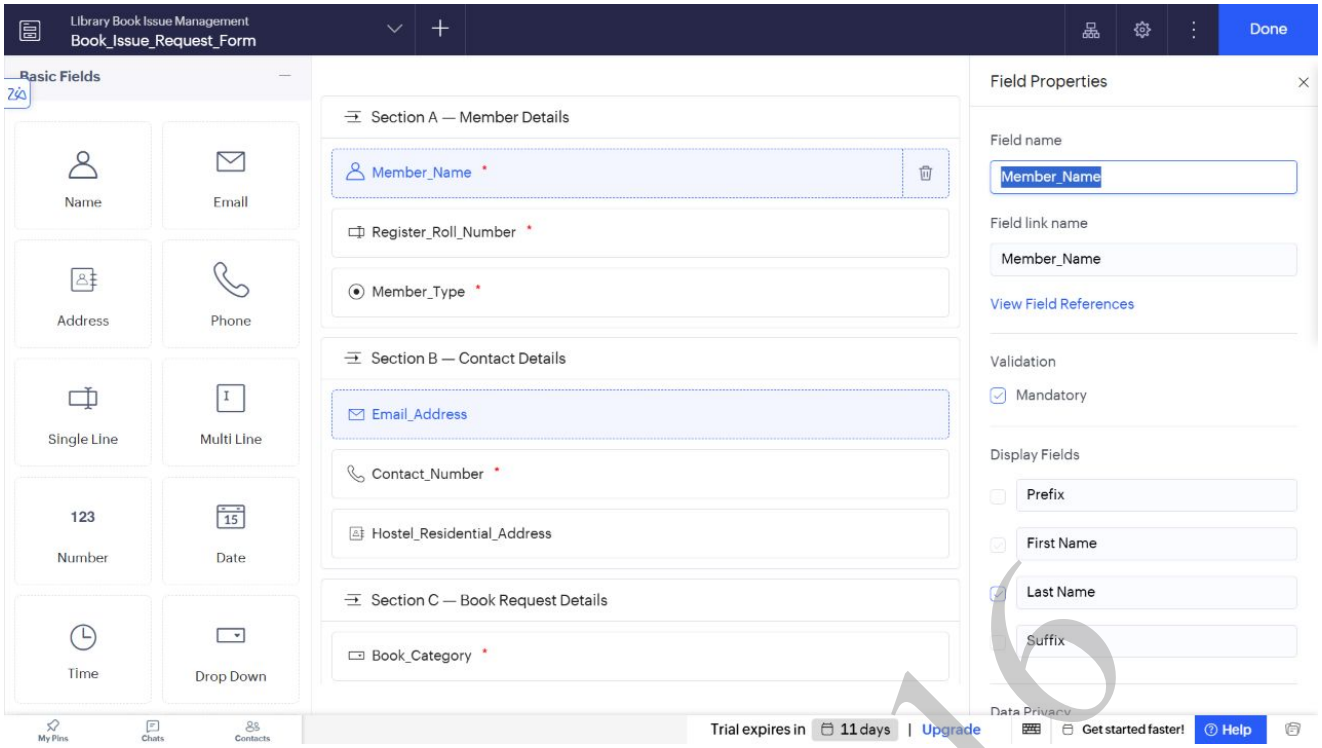


Figure 4.1. Development Form Builder: member and contact fields; Member_Name is mandatory.

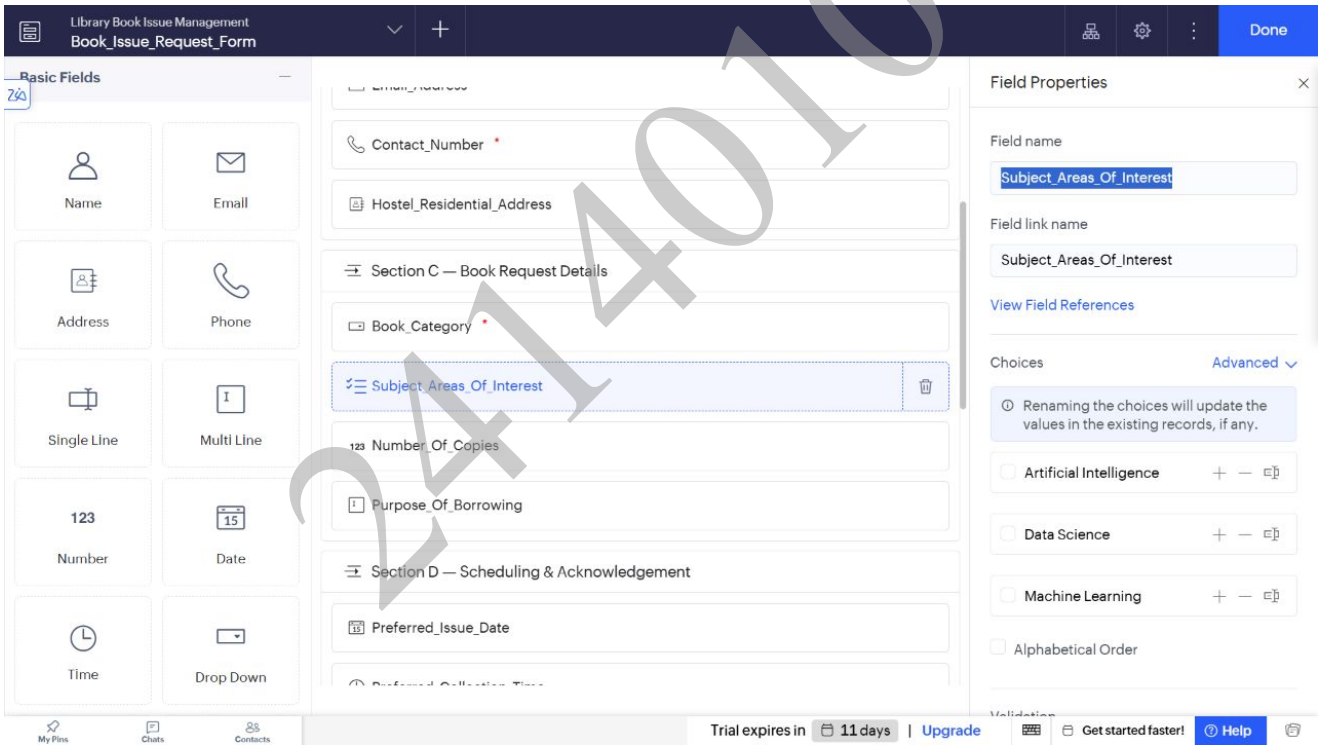


Figure 4.2. Development Form Builder: book-request fields and the three subject-area choices.

241401016

10 - Implementation Evidence

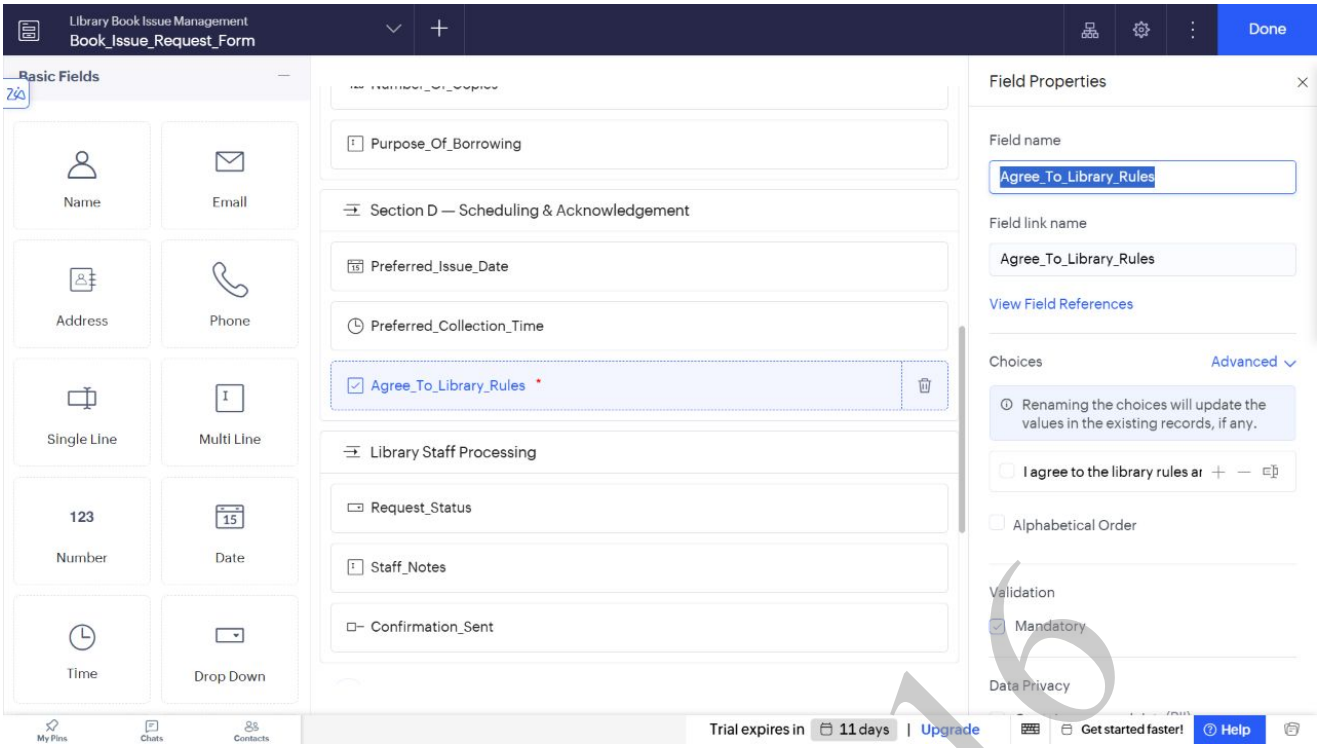


Figure 4.3. Development Form Builder: date, time, mandatory consent and additional staff processing fields.

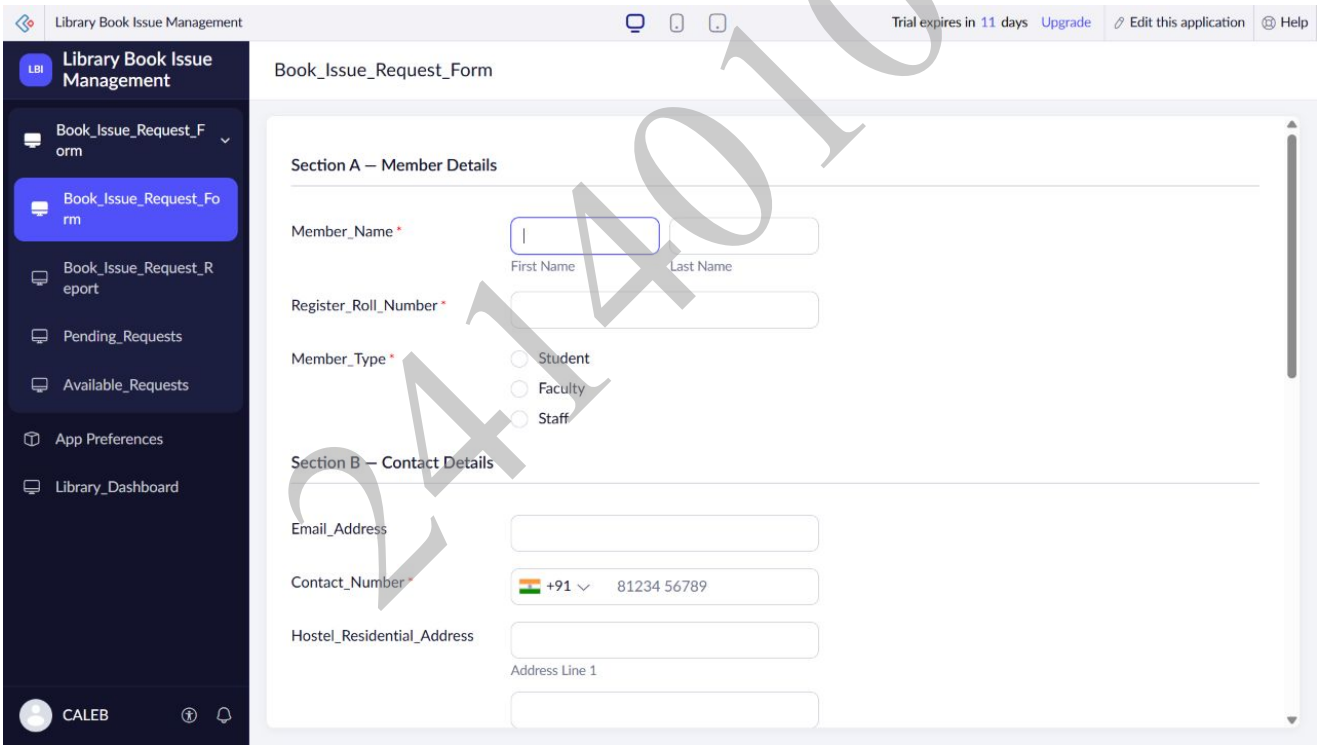


Figure 4.4. Production live form: Member Details and Contact Details with required-field markers.

241401016

10 - Implementation Evidence

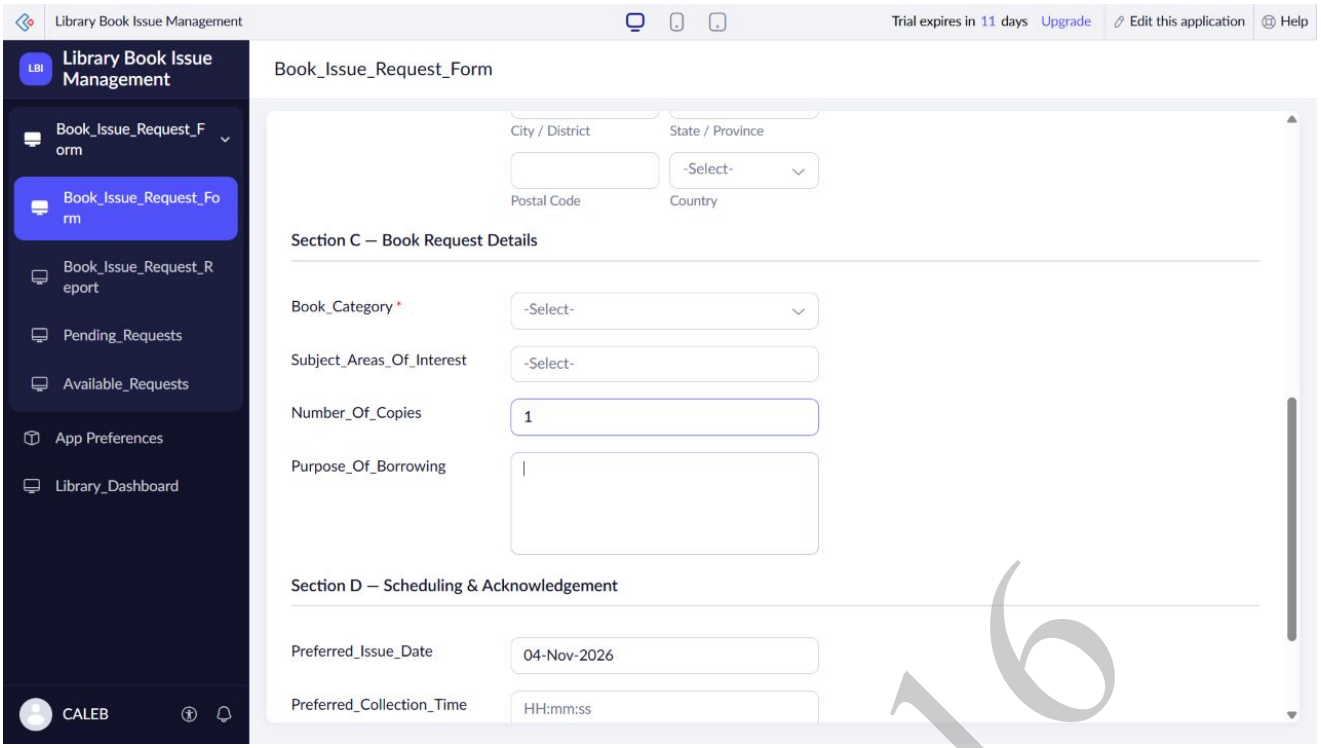


Figure 4.5. Production live form: book category, subjects, copies and borrowing purpose; unsaved preview.

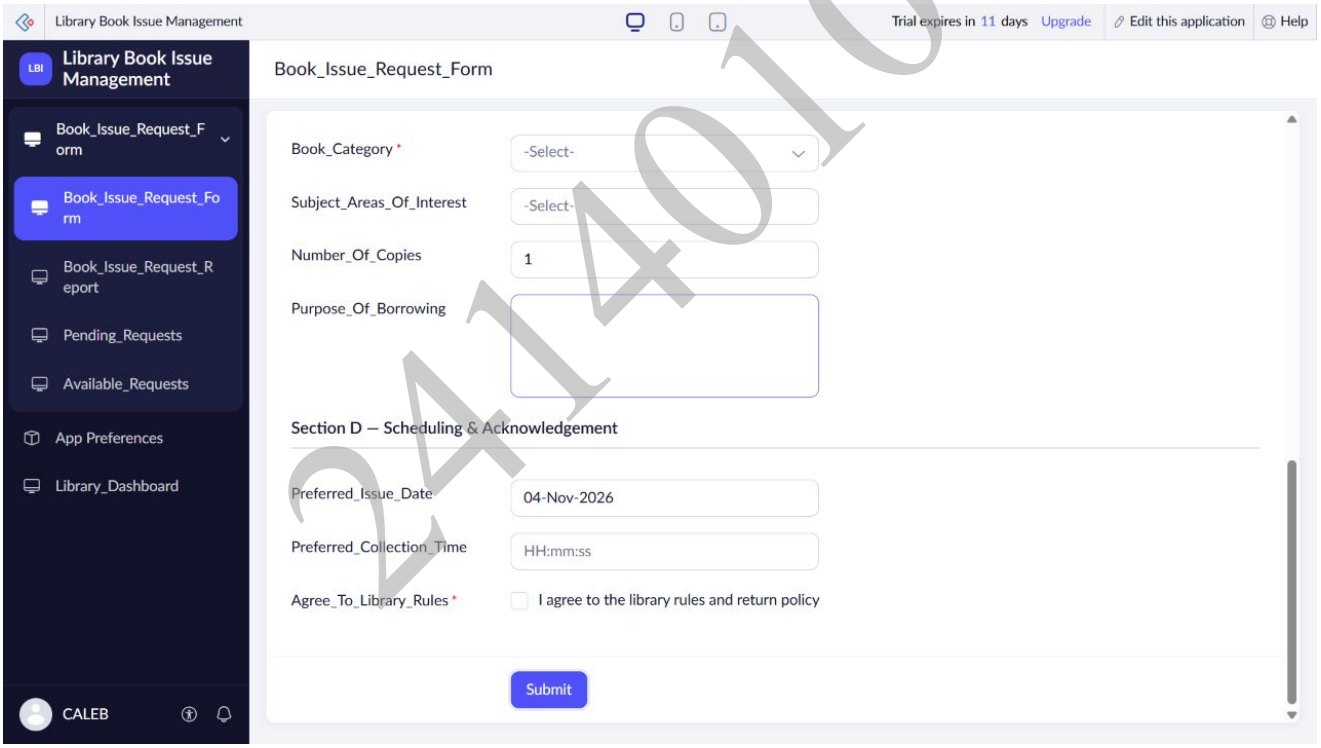


Figure 4.6. Production live form: scheduling, acknowledgement and Submit control; unsaved preview.

241401016

10 - Implementation Evidence

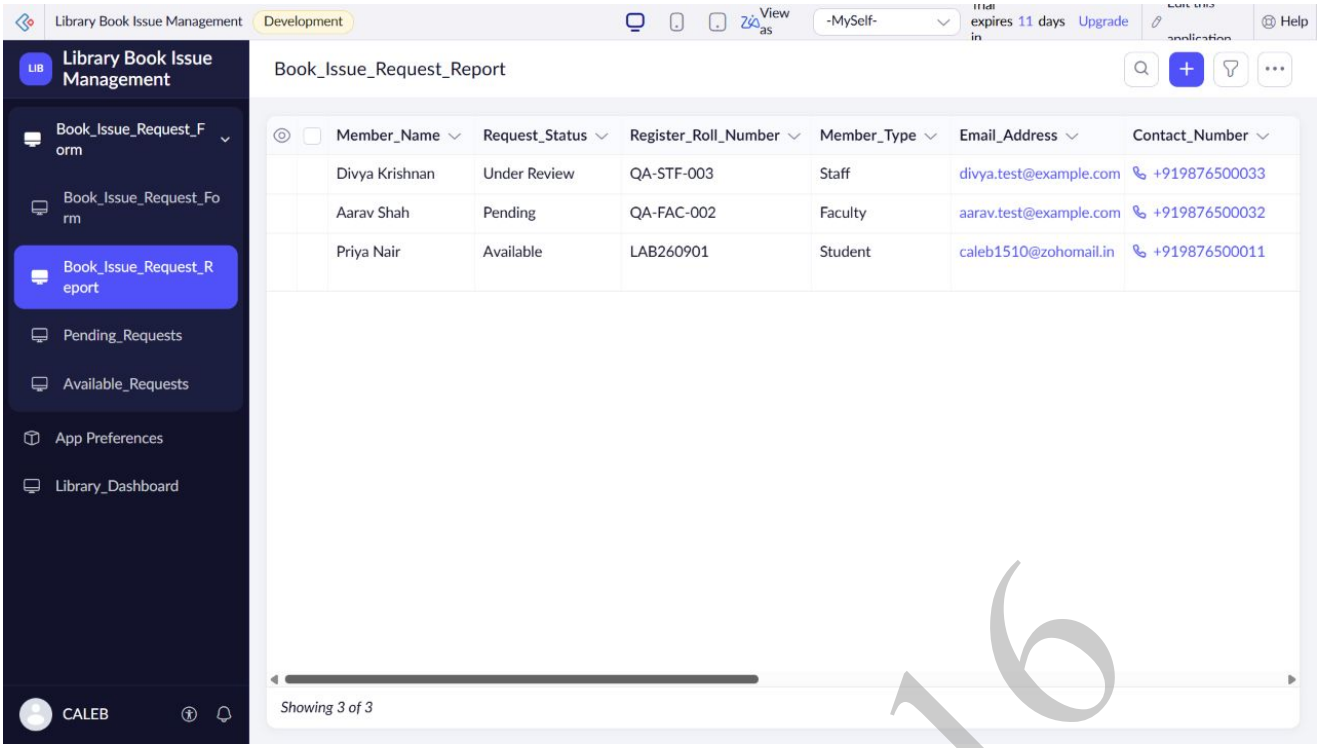


Figure 4.7. Development report with three submitted sample records: Priya, Aarav and Divya.

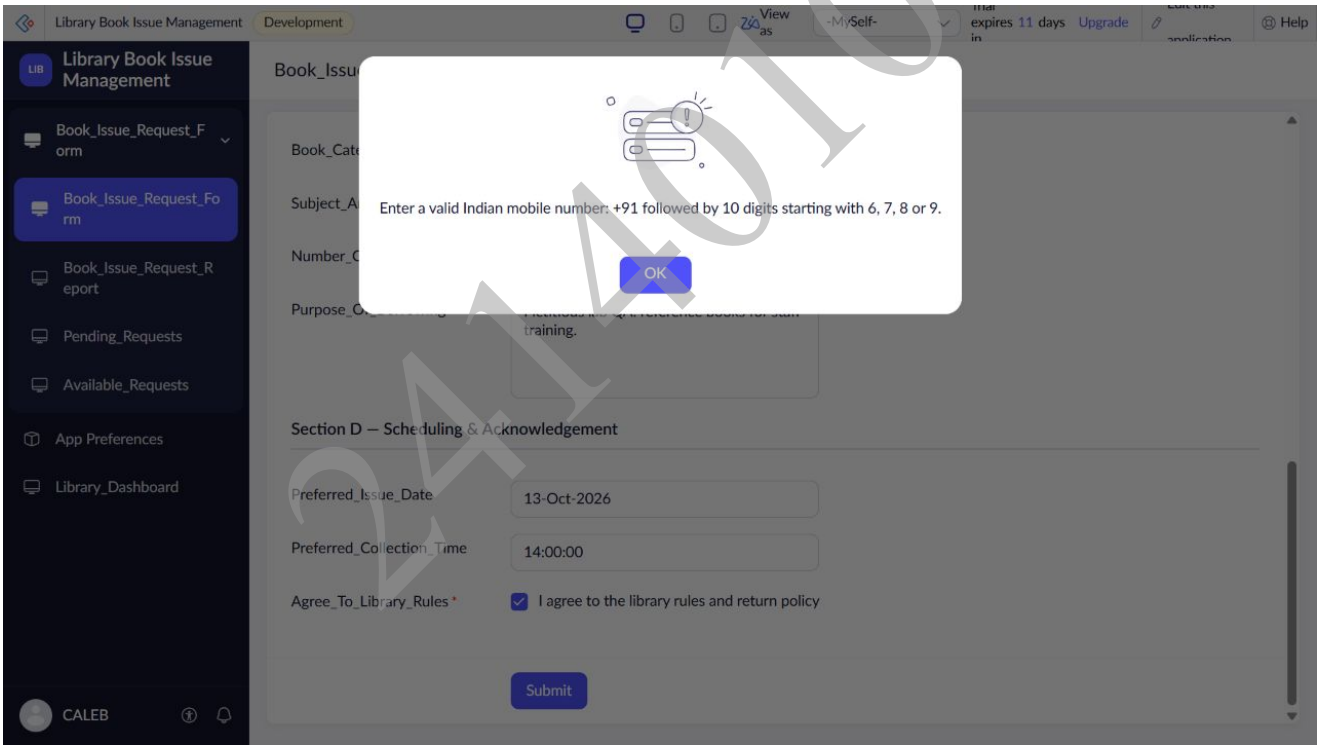


Figure 4.8. Development validation test rejects an invalid Indian mobile number.

241401016

11. Expected Learning Outcomes

- Map every Basic Field to a meaningful business attribute.

- Organize a form into four member-friendly sections.

- Apply mandatory, format, range, date and acknowledgement checks.

- Understand how submitted form values become report records.

12. Result

Book_Issue_Request_Form was implemented with all thirteen required fields and four sections. Mandatory and custom validations were configured, and more than three sample records were created across the tested environments.

Conclusion

The application replaces an unstructured library request register with typed, validated records that can be reviewed and processed by library staff.

Verification scope

The public form link was reported completed by the user; anonymous access was not independently retested during this documentation run. The evidence here verifies builder, authenticated form and saved-record behavior.

Rajalakshmi Engineering College

241401016